The effects of global warming

Compared to the start of the industrial revolution in 1750, carbon dioxide (CO_2) in the atmosphere has increased by about 50 percent. Alongside this, in 2024, the average global temperature was 2.7ºF (1.46ºC) higher than in 1750. Scientific evidence proves that these changes have contributed to melting glaciers and sea ice followed by rising sea levels—8 in (21 cm) since 1990—as well as damage to coral reefs. Other phenomena include longer wildfire seasons, more extreme weather, and shifts in the ranges of animals and plants, leading to disease, extinction, and food shortages.

The extent of the global temperature increase depends on whether (and how rapidly) global CO_2 emissions diminish. Scientists predict that, at the current rate, this increase could range from 0.5°F to 8°F (0.3 °–4.6 °C) by 2100, with the greatest warming likely in the Arctic region.